2023 WSMO Team Round Solutions

SMO Team

TR 1: Bob has a number of pencils. When Bob splits them into groups of 10, he has 3 left over. When he splits them into groups of 12, he has 5 left over. Find the smallest number of pencils Bob can have.

Answer: $\boxed{53}$

Solution: From the first condition, possible numbers of pencils Bob could have are

$$3, 13, 23, 33, 43, 53, 63, \ldots$$

and from the second condition we have

$$5, 17, 29, 41, 53, 65, \ldots.$$

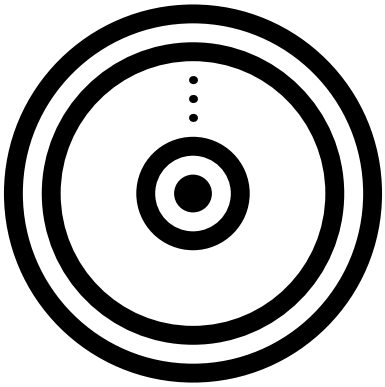
So, the smallest number of pencils Bob could have is $\boxed{53}$.

TR 2: Positive integers W, S, M, and O satisfy $W + S + M + O = 13$. Find the maximum possible value of $WSMO$.

Answer: $\boxed{108}$

Solution: To maximize the product of integers with a fixed sum, the numbers must be as close to each other as possible. So, let $W = S = M = 3$, and $O = 4$, which gives $WSMO = 3^3 \cdot 4 = \boxed{108}$.

TR 3: In the figure below, there are 1023 total circles with radii $1, 2, \ldots, 1023$. The area between circles alternates between shaded and non-shaded. The area of the shaded region can be expressed as $k\pi$. Find the remainder when k is divided by 1000.



Answer: $\boxed{776}$

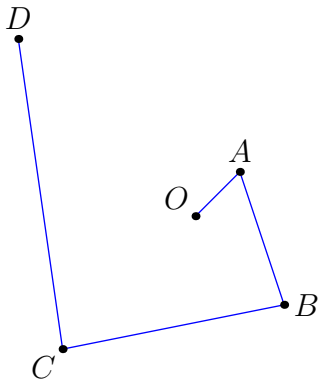
Solution: Note that

$$\begin{aligned} k &= 1023^2 - 1022^2 + 1021^2 - 1020^2 + \ldots + 3^2 - 2^2 + 1^2 \\ &= (1023 - 1022)(1023 + 1022) + \ldots + (3 - 2)(3 + 2) + 1^2 \\ &= (1023 + 1022) + (1021 + 1020) + \ldots + (3 + 2) + 1 \\ &= \frac{1023 \cdot 1024}{2} \implies \\ k &\equiv \boxed{776} \pmod{1000}. \end{aligned}$$

TR 4: Honko the hamster is in his cage. He wants to find the smallest distance needed to travel to reach four tennis balls. His current position is $(0, 0)$. The tennis balls are located at $(1, 1)$, $(2, -2)$, $(-3, -3)$, and $(-4, 4)$. The length of the shortest path can be expressed as $\sqrt{a_1} + \sqrt{a_2} + \cdots + \sqrt{a_n}$, where $a_1, \ldots, a_n$ are positive integers and n is as small as possible. Find $a_1 + a_2 + \cdots + a_n$.

Answer: $\boxed{108}$

Solution:



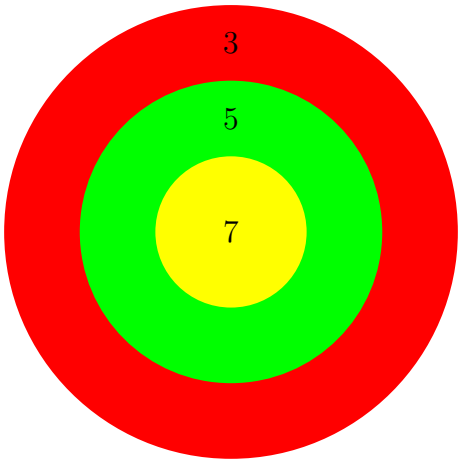
Let $O = (0, 0)$, $A = (1, 1)$, $B = (2, -2)$, $C = (-3, -3)$, and $D = (-4, 4)$. We compute $AB = \sqrt{10}$, $BC = \sqrt{26}$, $CD = \sqrt{50}$, and $DA = \sqrt{34}$. From casework on the first tennis ball Honko reaches, we have the following possible paths

$$\begin{aligned} &O \to A \to B \to C \to D \\ &O \to B \to A \to D \to C \\ &O \to C \to B \to A \to D \\ &O \to D \to A \to B \to C, \end{aligned}$$

of which the first one has the shortest length of

$$\sqrt{2} + \sqrt{10} + \sqrt{26} + \sqrt{50} = \sqrt{10} + \sqrt{26} + \sqrt{72} \implies 10 + 26 + 72 = \boxed{108}.$$

TR 5: A monkey is throwing darts at the dart board pictured below. The dart is equally likely to land anywhere on the board. Point values for the three regions are labeled and the radii of the three circles are 1, 2, and 3, respectively. The expected value of points the monkey gets from 5 dart throws can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.



Answer: $\boxed{194}$

Solution: The areas of the $3, 5, 7$ point regions are $5\pi, 3\pi, \pi$, respectively. So, the expected points of each throw is

$$\frac{5\pi}{9\pi} \cdot 3 + \frac{3\pi}{9\pi} \cdot 5 + \frac{\pi}{9\pi} \cdot 7 = \frac{37}{9}.$$

The expected points the monkey gets from 5 dart throws is

$$5 \cdot \frac{37}{9} = \frac{185}{9} \implies 185 + 9 = \boxed{194}.$$

TR 6: A quartic real polynomial $f(x)$ satisfying $f(3 + 2i) = 0$ has 3 distinct roots that sum to 12. Find the product of all the roots of f.

Answer: $\boxed{468}$

Solution: From the conjugate root theorem, since $3 + 2i$ is a root, $3 - 2i$ must also be a root. Since there are only three distinct roots, let x be the value of the remaining two roots. We have

$$x + (3 + 2i) + (3 - 2i) = 12 \implies x + 6 = 12 \implies x = 6.$$

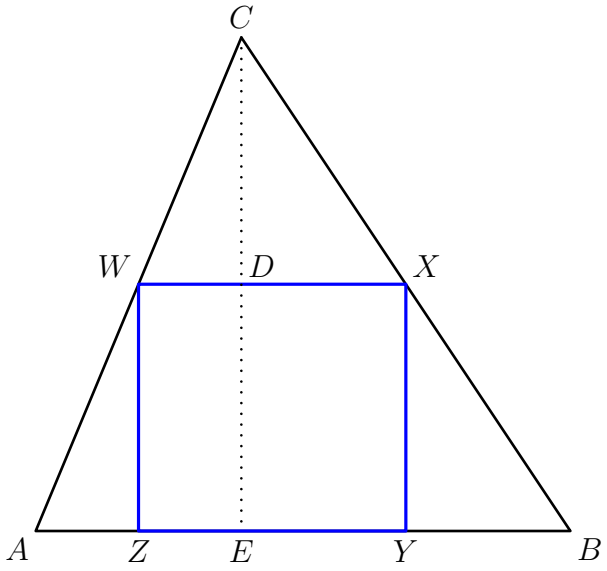
So, the product of the four roots is

$$(3 + 2i)(3 - 2i)(6)(6) = (3^2 + 2^2)(6)(6) = \boxed{468}.$$

TR 7: In triangle ABC, $AB = 13, AC = 14$, and $BC = 15$. A rectangle R is inscribed such that one side of R lies along a side of ABC and the other two vertices lie on the remaining two sides. Suppose the area of R is maximized. The minimum possible ratio of the length of the side of R lying along a side of ABC to its adjacent side length can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{337}$

Solution:



Note that $WXYZ$ must have two vertices on a side of ABC and the other two vertices on each of the other two sides of ABC. Assume WLOG that Z, Y lie on AB, W lies on AC, and X lies on CB. For simplicity, denote $c = AB = 13$ and $h = \frac{168}{c}$ the C-height of triangle ABC. Let $WX = ZY = x$ and $XY = WZ = y$. Let D and E be the feet of the perpendiculars from C to WX and AB, respectively. From similar triangles $\triangle CWX \sim \triangle CAB$, we have

$$\frac{CD}{CE} = \frac{WX}{AB} \implies \frac{h-y}{h} = \frac{x}{c} \implies \frac{\frac{168}{c} - y}{\frac{168}{c}} = \frac{x}{c} \implies x = \frac{168c - c^2 y}{168}.$$

So,

$$[WXYZ] = xy = \left(\frac{168c - c^2 y}{168}\right) y = \frac{168cy - c^2 y^2}{168},$$

which is maximized when $y = -\frac{168c}{-2c^2} = \frac{84}{c}$. Substituting, we have

$$x = \frac{168c - c^2 \cdot \frac{84}{c}}{168} = \frac{84c}{168} = \frac{c}{2},$$

meaning

$$\frac{WX}{XY} = \frac{x}{y} = \frac{\frac{c}{2}}{\frac{84}{c}} = \frac{c^2}{168} = \frac{169}{168} \implies 169 + 168 = \boxed{337}.$$

Note that if $WXYZ$ had two vertices on sides AC, BC, then we still have $[WXYZ]$ maximized to $\frac{[ABC]}{2}$ but $\frac{WX}{XY} = \frac{14^2}{168}, \frac{15^2}{168} > \frac{169}{168}$, respectively.

TR 8: Let $f(x) = \sqrt{x - \sqrt{x - \sqrt{x - \ldots}}}$. Find the remainder when the least integer a such that $f(f(f(f(f(a)))))$ is a positive integer is divided by 1000.

Answer: $\boxed{442}$

Solution: Note that

$$\begin{aligned} f(x) = \sqrt{x - f(x)} &\implies \\ [f(x)]^2 = x - f(x) &\implies \\ x = [f(x)]^2 + f(x). & \end{aligned}$$

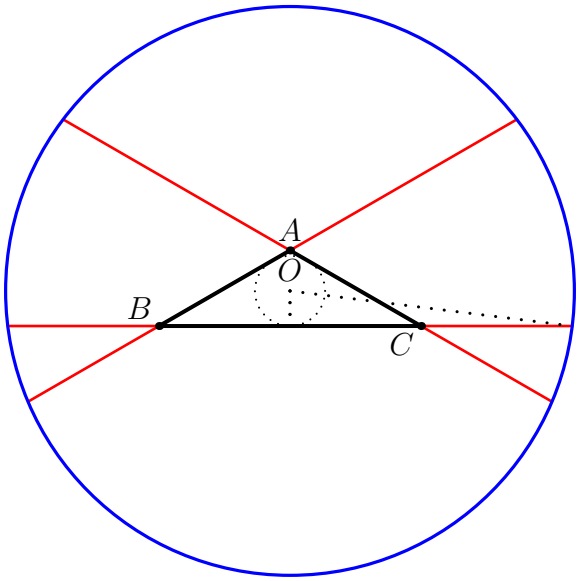
Since $[f(x)]^2 + f(x)$ is a strictly increasing function, the minimum value of a occurs when

$$\begin{aligned} f(f(f(f(f(a))))) &= 1 \implies \\ f(f(f(f(a)))) &= 1^2 + 1 = 2 \implies \\ f(f(f(a))) &= 2^2 + 2 = 6 \implies \\ f(f(a)) &= 6^2 + 6 = 42 \implies \\ f(a) &= 42^2 + 42 = 1806 \implies \\ a &= 1806^2 + 1806 \implies \\ a &\equiv \boxed{442} \pmod{1000}. \end{aligned}$$

TR 9: A circle ω has three chords of equal length, $4 + 2\sqrt{3}$, which intersect forming a triangle with side lengths 2, 2, and $2\sqrt{3}$. The square of the radius of ω can be expressed as $a - b\sqrt{c}$, where a, b, and c are positive integers and c is not divisible by the square of any prime. Find $a + b + c$.

Answer: $\boxed{39}$

Solution:



Let the triangle formed from the intersection be ABC and the center of the circle be O. Since the three chords are of equal length, O must be equidistant to sides AB, BC, and AC, meaning O is the incenter of ABC. The inradius is equal to the quotient of the area and semiperimeter, which is

$$\frac{\sqrt{3}}{\frac{2+2+2\sqrt{3}}{2}} = \frac{\sqrt{3}}{2+\sqrt{3}} = 2\sqrt{3} - 3.$$

From the Pythagorean Theorem, the square of the radius of ω is

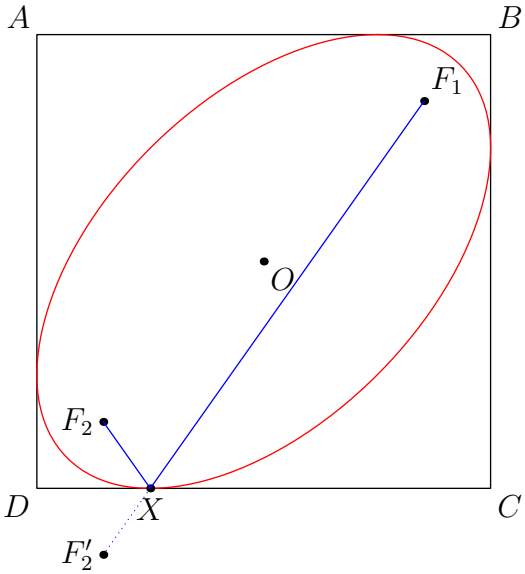
$$(2\sqrt{3} - 3)^2 + (2 + \sqrt{3})^2 = (21 - 12\sqrt{3}) + (7 + 4\sqrt{3}) = 28 - 8\sqrt{3} \implies 28 + 8 + 3 = \boxed{39}.$$

TR 10: Let square $ABCD$ be a square with side length 4. Define ellipse ω as the ellipse that is able to be inscribed inside $ABCD$ such that 2 of its vertices on its minor axis and 1

of its vertices on its major axis form an equilateral triangle. The largest possible area of ω can be expressed as $m\pi\sqrt{n}$, where m and n are positive integers and n is not divisible by the square of any prime. Find $m+n$.

Answer: $\boxed{5}$

Solution:



We will use analytic geometry. Let $O = (0,0)$ be the center of the square, giving vertices $A = (-2,2), B = (2,2), C = (2,-2)$, and $D = (-2,-2)$.

Let the side length of the equilateral triangle be $2t$. The semi-minor axis is $b = t$ and the semi-major axis is $a = t\sqrt{3}$, so the area is $\pi ab = \pi t^2\sqrt{3}$. Maximizing the area is equivalent to maximizing t. For any fixed orientation θ, scaling the ellipse up until it is tangent to the sides of the square maximizes t, so the maximal ellipse must touch the sides. Furthermore, an ellipse centered at O has bounding box half-widths $\sqrt{a^2\cos^2\theta + b^2\sin^2\theta} \leq 2$ and $\sqrt{a^2\sin^2\theta + b^2\cos^2\theta} \leq 2$. Adding the squares of these bounds, we have $a^2 + b^2 \leq 8$, with equality only when $\theta = 45°$.

Hence, the major axis lies along $y = x$ $(45°)$, meaning the ellipse is centered at O and touches all four sides. The focal distance is $2c = 2\sqrt{a^2 - b^2} = 2t\sqrt{2}$, so the foci are $F_1 = (t,t)$ and $F_2 = (-t,-t)$. By definition, any point P on the ellipse satisfies $F_1P + F_2P = 2a = 2t\sqrt{3}$.

Let X be the tangency point of the ellipse with CD and F_2' be the reflection of F_2 across CD. Due to tangency, $F_1P + F_2P$ is minimized at $P = X$ over all points P on CD. Since CD is the perpendicular bisector of F_2F_2', $F_1X + F_2X = F_1X + XF_2'$. From the triangle inequality, X must be located on F_1F_2'. So, $F_1X + XF_2' = F_1F_2'$. Since F_2' is the reflection of $F_2 = (-t,-t)$ across $CD : y = -2$, we have $F_2' = (-t, -4+t)$. So, $(F_1F_2')^2 = (2t)^2 + (-4)^2 = 4t^2 + 16$. Setting this equal to $(2t\sqrt{3})^2$, we have $4t^2 + 16 = 12t^2 \implies t^2 = 2$. This means the major and minor axes have length $2\sqrt{6}, 2\sqrt{2}$, respectively, meaning our answer is $(\sqrt{6})(\sqrt{2})\pi = 2\pi\sqrt{3} \implies 2+3 = \boxed{5}$.

TR 11: When $\frac{1}{7}$ is expressed in base k, the digits are $0.a_{k,1}a_{k,2}\ldots$, where $a_{k,1}, a_{k,2}, \ldots$ are base-k digits. Let $f(p)$ denote the minimum positive integer $x \geq 2$ such that $a_{p,1} = a_{p,x}$, for $p \not\equiv 0 \pmod 7$, and 1 for $p \equiv 0 \pmod 7$. Find $\sum_{i=2}^{100} f(i)$.

Answer: $\boxed{390}$

Solution: We have

$$0.a_{p,1}a_{p,2}\cdots = 0.\overline{a_{p,1}a_{p,2}\ldots a_{p,f(p)-1}} = \frac{\overline{a_{p,1}a_{p,2}\ldots a_{p,f(p)-1}}_p}{p^{f(p)-1}-1}.$$

For $p \geq 7$, $f(p)-1$ is the order of p modulo 7. From here, we compute

$$f(p) = 1, 2, 4, 7, 4, 7, 3 \qquad \text{for } p \equiv 0, 1, 2, 3, 4, 5, 6 \pmod 7,$$

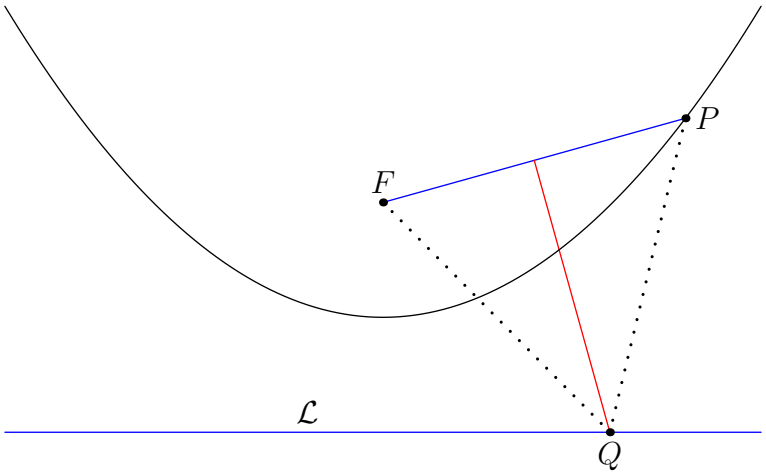
respectively. However, for $p < 7$, checking directly gives $f(2) = 2$ (instead of 4) and $f(3) = 3$ (instead of 7). So,

$$\sum_{i=2}^{100} f(i) = (1+2+4+7+4+7+3)\cdot 14 + 4 - (4-2) - (7-3) = \boxed{390}.$$

TR 12: Consider parabola $\mathcal{P}$ pointing upwards with vertex at the origin of the Cartesian plane. Denote the focus of it as F and the directrix of it as $\mathcal{L}$. Point P with x-coordinate 4 is selected on $\mathcal{P}$. The perpendicular bisector of FP meets $\mathcal{L}$ at Q. Given the x-coordinate of Q is 3, then FP^2 can be expressed as $\frac{a+b\sqrt{c}}{d}$, where a, b, c, and d are positive integers, c is not divisible by the square of any prime, and $\gcd(a, b, d) = 1$. Find $a+b+c+d$.

Answer: $\boxed{46}$

Solution:



Let the parabola have equation $y = kx^2$. We have

$$F = \left(0, \frac{1}{4k}\right) \quad \text{and} \quad \mathcal{L}: y = -\frac{1}{4k}.$$

Since the x-coordinate of P is 4, we have $P = (4, 16k)$. Since Q lies on $\mathcal{L}$ and has x-coordinate 3, we have $Q = \left(3, -\frac{1}{4k}\right)$. Since Q lies on the perpendicular bisector of FP, we

have $FQ = PQ$. So

$$(3-0)^2 + \left(\frac{1}{4k} - \left(-\frac{1}{4k}\right)\right)^2 = (4-3)^2 + \left(16k + \frac{1}{4k}\right)^2 \implies$$
$$3^2 + \frac{1}{4k^2} = 1 + \left(256k^2 + 8 + \frac{1}{16k^2}\right) \implies$$
$$9 + \frac{1}{4k^2} = 9 + 256k^2 + \frac{1}{16k^2} \implies$$
$$\frac{1}{4k^2} = 256k^2 + \frac{1}{16k^2} \implies$$
$$\frac{3}{16k^2} = 256k^2 \implies k^4 = \frac{3}{4096} \implies k = \frac{\sqrt[4]{3}}{8}.$$

Now,

$$\begin{aligned} FP^2 &= 4^2 + \left(16k - \frac{1}{4k}\right)^2 \\ &= 16 + \frac{(64k^2-1)^2}{16k^2} \\ &= 16 + \frac{(\sqrt{3}-1)^2}{\frac{\sqrt{3}}{4}} \\ &= \frac{24+16\sqrt{3}}{3} \implies 24+16+3+3 = \boxed{46}. \end{aligned}$$

TR 13: Gerry the Squirrel is at a corner of a 9×9 grid of points. An acorn is placed randomly on a different point of the grid. Four coins are placed randomly on the grid, none of which are located on Gerry's starting point or on the acorn. Gerry will take a shortest path to the acorn, and if there are multiple shortest paths, Gerry will pick one randomly. The expected number of coins Gerry passes during his trip can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Answer: $\boxed{537}$

Solution: Place Gerry at $(0,0)$. Suppose the acorn is located at (x,y), where $0 \le x, y \le 8, (x,y) \neq (0,0)$ (so 80 possible positions). Any shortest path from $(0,0)$ to (x,y) passes through $x+y-1$ lattice points. So, the expected number of points Gerry passes through is

$$\frac{\sum_{x=0}^{8}\sum_{y=0}^{8}(x+y-1) - (0+0-1)}{80} = \frac{81 \cdot (4+4-1)+1}{80} = \frac{568}{80}.$$

The four coins are placed randomly among the 79 remaining lattice points. Hence, each point Gerry passes has a $\frac{4}{79}$ probability of containing a coin. So, the expected number of coins Gerry passes is

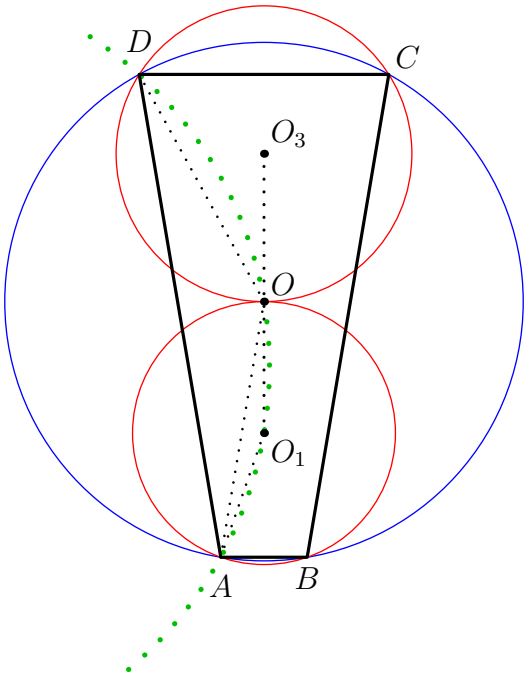
$$\frac{568}{80} \cdot \frac{4}{79} = \frac{142}{395} \implies 142 + 395 = \boxed{537}.$$

TR 14: Consider quadrilateral $ABCD$ with circumcircle centered at O. Given $AB = 2 < CD$, the circumcircle of $\triangle ABO$ is tangent to the circumcircle of $\triangle COD$. The circumcircle of $\triangle AOD$ passes through the center of the circumcircle of $\triangle ABO$. The radius of the circumcircle of $ABCD$ is 6. Let M and N be the area and perimeter of $ABCD$, respectively. The value of $\frac{M}{N}$ can be expressed as $\frac{a\sqrt{b}}{c}$, where a, b, and c are positive integers,

a and c are relatively prime, and b is not divisible by the square of any prime. Find $a+b+c$.

Answer: 3096

Solution:



Let O_3 and O_1 be the circumcenters of COD and ABO, respectively. Since ABO and COD share point O, they must be externally tangent at O, meaning O_3, O, and O_1 are collinear. Since $AO = BO$ and $CO = DO$, ABO and COD are isosceles, meaning $OO_1 \perp AB$ and $OO_3 \perp CD$. Since O_3, O, O_1 are collinear, $AB \parallel CD$, meaning $ABCD$ is an isosceles trapezoid. Now, the O-height of ABO is of length $\sqrt{6^2 - \left(\frac{2}{2}\right)^2} = \sqrt{35}$. So,

$$OO_1 = \frac{(OA)(AB)(BO)}{4[ABO]} = \frac{6 \cdot 2 \cdot 6}{4 \cdot \frac{2 \cdot \sqrt{35}}{2}} = \frac{18}{\sqrt{35}}.$$

Let $\theta = \angle OAO_1 = \angle AOO_1$. Since OO_1A is isosceles, we have

$$\cos\theta = \frac{\frac{OA}{2}}{O_1A} = \frac{3}{\frac{18}{\sqrt{35}}} = \frac{\sqrt{35}}{6} \implies \sin\theta = \sqrt{1 - \left(\frac{\sqrt{35}}{6}\right)^2} = \frac{1}{6}.$$

Now, from angle chasing, we have

$$\begin{aligned}
\angle OO_1A &= 180^\circ - 2\theta && \text{since } OO_1A \text{ is isosceles,} \\
\angle ODA &= 180^\circ - \angle OO_1A = 2\theta && \text{since } ODAO_1 \text{ is cyclic,} \\
\angle OAD &= \angle ODA = 2\theta && \text{since } AOD \text{ is isosceles,} \\
\angle AOB &= 2\angle AOO_1 = 2\theta && \text{from symmetry,} \\
\angle OAB &= \frac{180^\circ - \angle AOB}{2} = 90^\circ - \theta && \text{since } ABO \text{ is isosceles,} \\
\angle DAB = \angle DAO + \angle OAB &= 2\theta + (90^\circ - \theta) = 90^\circ + \theta && \text{, and} \\
\angle CDA &= 180^\circ - \angle DAB = 90^\circ - \theta && \text{since } AB \parallel CD.
\end{aligned}$$

Now,

$$\sin\angle OAD = \sin(2\theta) = 2\sin\theta\cos\theta = 2\left(\frac{1}{6}\right)\left(\frac{\sqrt{35}}{6}\right) = \frac{\sqrt{35}}{18},$$

so

$$\cos(2\theta) = \sqrt{1-\left(\frac{\sqrt{35}}{18}\right)^2} = \frac{17}{18}.$$

We have

$$AD = 2\cdot(AO\cos\angle OAD) = 12\cdot\cos(2\theta) = 12\cdot\frac{17}{18} = \frac{34}{3}.$$

Since $ABCD$ is an isosceles trapezoid, this means $CB = \frac{34}{3}$ as well. Now,

$$\begin{aligned} CD &= AB + 2AD\cos(\angle CDA) = 2 + 2AD\cos(90-\theta) \\ &= 2 + 2AD\sin\theta = 2 + 2\left(\frac{34}{3}\right)\left(\frac{1}{6}\right) = \frac{52}{9}. \end{aligned}$$

This gives us a perimeter of

$$N = AB + BC + CD + DA = 2 + \frac{34}{3} + \frac{52}{9} + \frac{34}{3} = \frac{274}{9}$$

and an area of

$$\begin{aligned} M &= \left(\frac{AB+CD}{2}\right)(AD\sin\angle CDA) \\ &= \left(\frac{2+\frac{52}{9}}{2}\right)\left(\frac{34}{3}\sin(90-\theta)\right) \\ &= \frac{35}{9}\cdot\frac{34}{3}\cdot\cos\theta \\ &= \frac{35}{9}\cdot\frac{34}{3}\cdot\frac{\sqrt{35}}{6} \\ &= \frac{595\sqrt{35}}{81}. \end{aligned}$$

So,

$$\frac{M}{N} = \frac{\frac{595\sqrt{35}}{81}}{\frac{274}{9}} = \frac{595\sqrt{35}}{2466} \implies 595+35+2466 = \boxed{3096}.$$

TR 15: On a number line labeled 0, 1, 2, 3, 4, 5, an old man starts at 0 and tries to reach 5. Initially, he knows to walk right. However, he has dementia. On each move, there is a $\frac{1}{3}$ chance he forgets which direction he is supposed to go, resulting in him walking the opposite direction. If the old man tries to walk left when he is at 0, he falls off a cliff and dies. The probability the old man reaches 5 without dying can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Answer: $\boxed{95}$

Solution: Let P_x be the probability the old man reaches 5. We have

$$P_0 = \frac{2}{3}P_1 \implies P_1 = \frac{3}{2}P_0,$$
$$P_1 = \frac{2}{3}P_2 + \frac{1}{3}P_0 \implies \frac{3}{2}P_0 = \frac{2}{3}P_2 + \frac{1}{3}P_0 \implies$$
$$\frac{7}{6}P_0 = \frac{2}{3}P_2 \implies P_2 = \frac{7}{4}P_0$$
$$P_2 = \frac{2}{3}P_3 + \frac{1}{3}P_1 \implies \frac{7}{4}P_0 = \frac{2}{3}P_3 + \frac{1}{2}P_0 \implies$$
$$\frac{5}{4}P_0 = \frac{2}{3}P_3 \implies P_3 = \frac{15}{8}P_0$$
$$P_3 = \frac{2}{3}P_4 + \frac{1}{3}P_2 \implies \frac{15}{8}P_0 = \frac{2}{3}P_4 + \frac{7}{12}P_0 \implies$$
$$\frac{31}{24}P_0 = \frac{2}{3}P_4 \implies P_4 = \frac{31}{16}P_0$$
$$P_4 = \frac{2}{3} + \frac{1}{3}P_3 \implies \frac{31}{16}P_0 = \frac{2}{3} + \frac{5}{8}P_0 \implies$$
$$\frac{21}{16}P_0 = \frac{2}{3} \implies P_0 = \frac{32}{63} \implies 32 + 63 = \boxed{95}.$$